Stochastic Processes 2021/22 Semester 1

Tutorial 4

1. Consider again the Markov chain $\{X_n\}_{n\geq 0}$ of *Tutorial 3, Q1*, taking values in the state space $S = \{0, 1, 2\}$ with transition matrix

$$P = \begin{pmatrix} 1/2 & 0 & 1/2 \\ 1 & 0 & 0 \\ 1/4 & 1/2 & 1/4 \end{pmatrix}.$$

   (a) Verify that all states form a single, necessarily closed, class.

   (b) Find the unique stationary distribution for the chain.

   (c) Suppose again that $\mathbf{P}(X_0 = i) = 1/3$ for $i = 0, 1, 2$. Find (again) the distributions of the random variables X_1, X_2, X_3 and comment on the rate of converge to the stationary distribution.

2. Consider again the taxicab driver of *Tutorial 3, Q2*.

   (a) Find the stationary distribution of his location.

   (b) In the long term, what proportion of his journeys terminate at a hotel?

3. Consider again the two-state Markov chain of *Tutorial 3, Q3*. Find the stationary distribution $\boldsymbol{\pi}$ associated with this chain and verify that, for all i, j, we have $\lim_{n\to\infty} p_{ij}^{(n)} = \pi_j$.

4. The NCD scheme operated by a motor insurer has three levels of discount, $0\%, 30\%$ and 50%. The rules for moving between discount levels are:

   > If a driver makes no claims in a year (s)he moves in the following year to the next higher rate of discount or stays at 50%.
   >
   > If a driver makes 1 claim in a year (s)he moves in the following year to the next lower discount level or stays at 0%.
   >
   > If a driver makes 2 or more claims, (s)he moves in the following year to the 0% discount level.

   The probability of one claim in a year for any individual in a certain risk group is 0.14, and the probability of 2 or more claims is 0.01.

   (a) Calculate the transition matrix of the Markov chain modelling the movements of any one individual in the risk group.

   (b) Calculate the expected proportion of this risk group at each discount level once a stationary position is reached.

5. A no claims discount system under a motor policy has four levels of discount: $0\%, 25\%, 40\%$ and 60%. The rules for moving between these levels are as follows:

   If the policyholder has no claims in a year, (s)he moves in the next year to the next higher level of discount, or remains at 60% discount.

   If the policyholder has one claim in a year, (s)he stays at the same level of discount in the next year.

   If the policyholder has two or more claims in a year, (s)he moves to 0% discount in the next year.

   The number of claims each year for each policyholder is assumed to have a Poisson distribution with parameter 0.2.

   (a) Find the transition matrix P of the process.

   (b) For $10,000$ policyholders at 0% discount at time 0, estimate the expected numbers at each discount level at times 1 and 2, assuming no exits.

   (c) For $10,000$ policyholders, estimate the expected numbers at each discount level once stability has been achieved, again assuming no exits.

6. A impoverished lecturer has 4 shirts, labelled $1, 2, 3, 4$. Each day he chooses at random a shirt to wear from those he was not wearing the previous day. Let X_n be the label of the shirt he wears on day n.

   (a) Explain why $\{X_n\}_{n\geq 1}$ is a Markov chain.

   (b) Show that the transition matrix P may be written

   $$P = \frac{4}{3}E - \frac{1}{3}I$$

   where E is the 4×4 matrix with every element equal to $\frac{1}{4}$ and I is the 4×4 identity matrix.

   (c) Prove, by induction on n, that

   $$P^n = \left(1 - \left(-\frac{1}{3}\right)^n\right) E + \left(-\frac{1}{3}\right)^n I$$

   (d) Find the stationary distribution $\boldsymbol{\pi}$ of the chain, and verify that $\lim_{n\to\infty} p_{ij}^{(n)} = \pi_j$ for all i, j.

   (e) Let $f_{ii}(n)$ be the probability that the next time he wears shirt i is in n days time, given that he is wearing it today. Show that

   $$f_{ii}(n) = \begin{cases} 0 & \text{for } n = 1 \\ \left(\frac{2}{3}\right)^{n-2} \frac{1}{3} & \text{for } n \geq 2. \end{cases}$$

   Hence show that the mean number of days between successive wearings of the shirt i is 4.

7. A finite non-negative $N \times N$ matrix is called *doubly stochastic* if each of its rows and each of its columns sum to 1. Show that if a Markov chain has a doubly stochastic transition matrix, then the distribution $N^{-1}(1, 1, \ldots, 1)$ is a stationary distribution for the chain. [*Note the application to the preceding question.*]

8. Two urns each contain N balls. Of the total of $2N$ balls, N are black and N are white. From time to time a ball is chosen at random from urn 1 and swapped with a ball chosen at random from urn 2. Let X_n denote the number of black balls in urn 1 after the nth swap.

   (a) Write down the transition matrix for the Markov chain $\{X_n\}$.

   (b) Verify that the stationary distribution for this chain is given by $\boldsymbol{\pi} = (\pi_0, \pi_1, \ldots, \pi_N)$, where

$$\pi_j = \frac{\binom{N}{j}^2}{\binom{2N}{N}}$$

   [*Recall that, by uniqueness, it is sufficient to verify that* $\boldsymbol{\pi}$ *as given above is indeed a distribution and does satisfy the defining equations for stationarity.*]

9. Consider further the simple random walk on the nonnegative integers, and modified to have a reflecting barrier at 0—as described in *Section 2.1.2* of the *Lecture Notes*. Show that in the case $p < 1/2$, this process has a stationary distribution $\boldsymbol{\pi}$ given by

$$\pi_i = (1 - p/q)\,(p/q)^i, \qquad i = 0, 1, 2, \ldots.$$